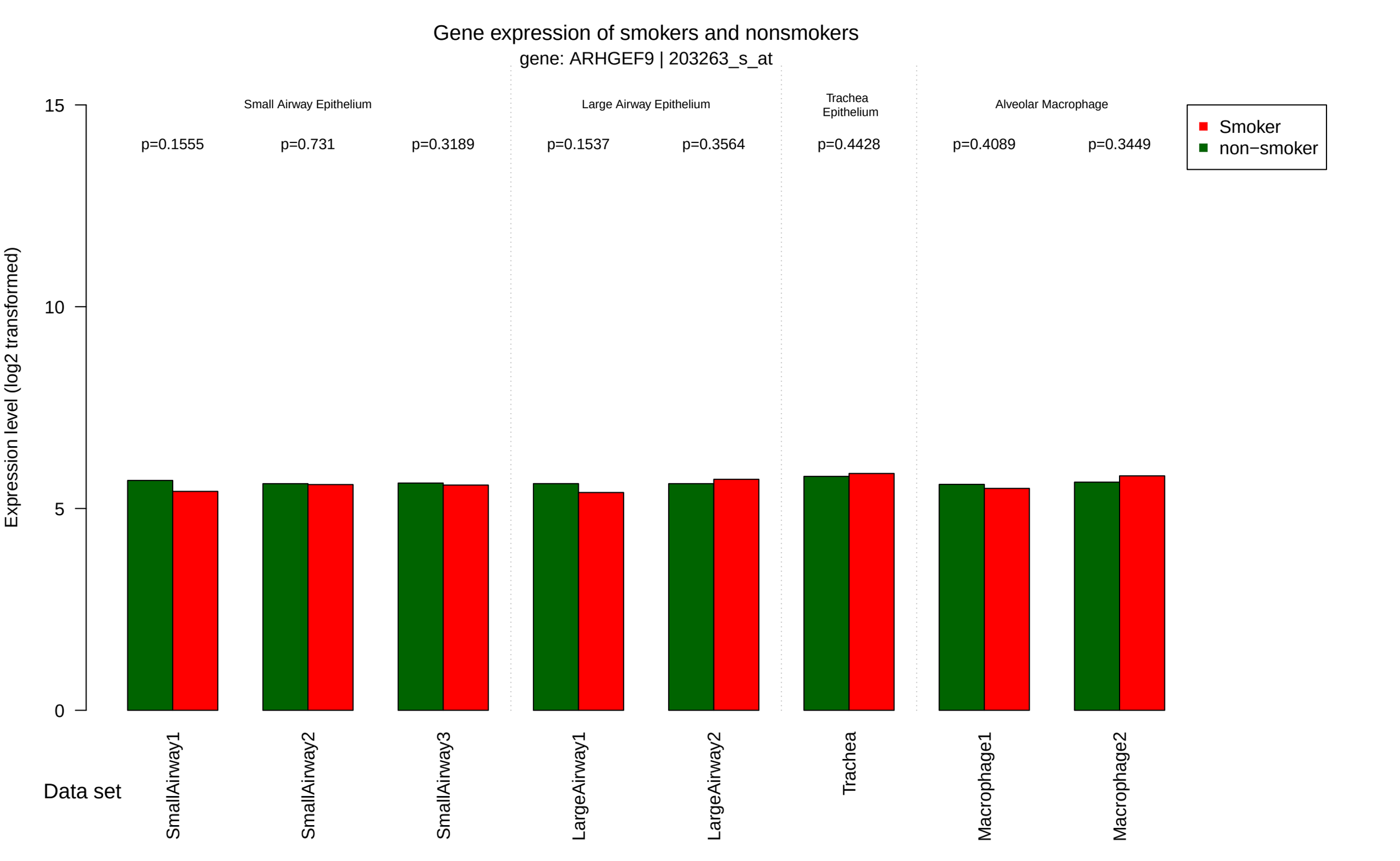
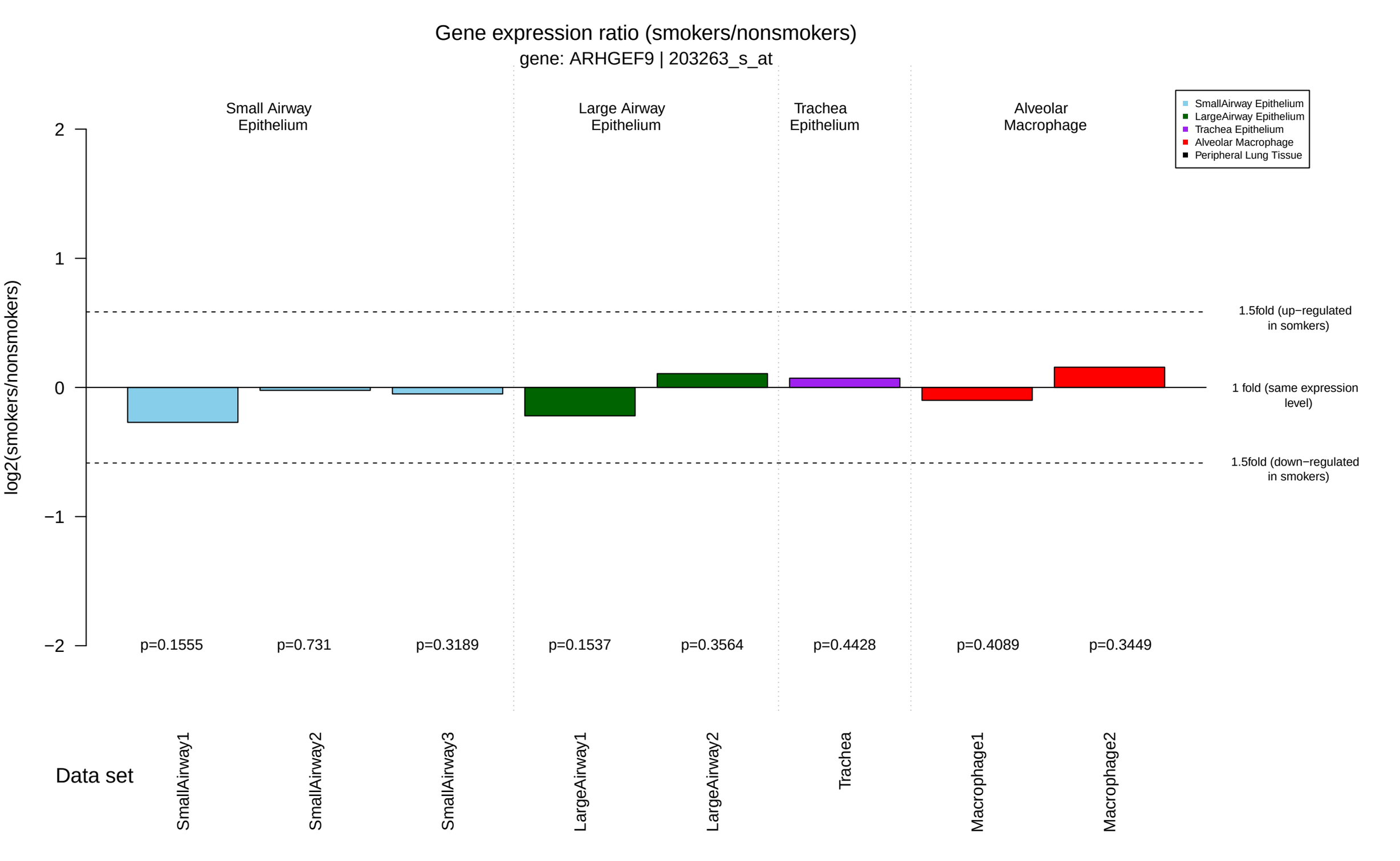
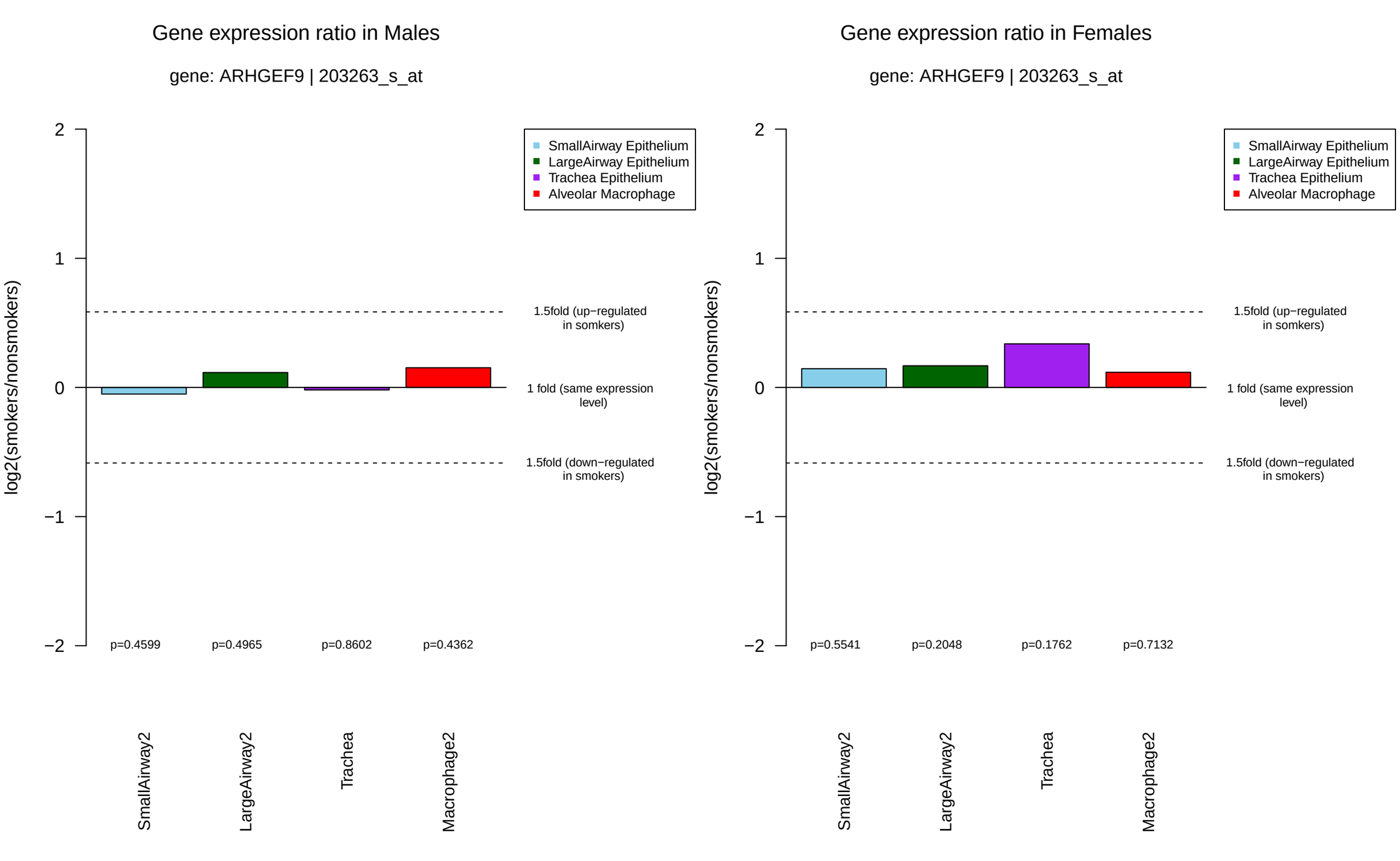
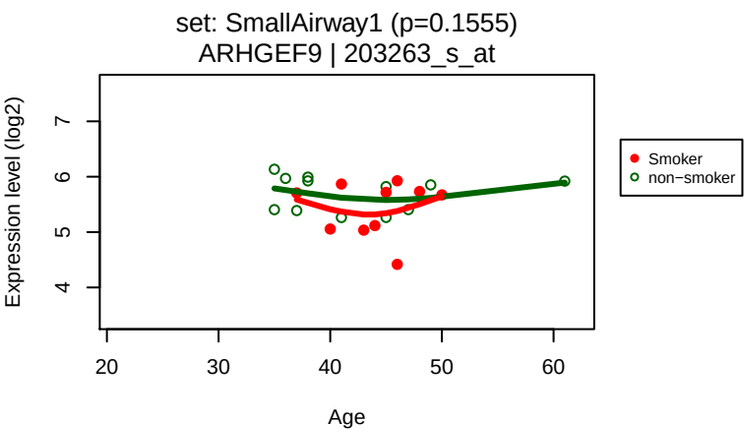
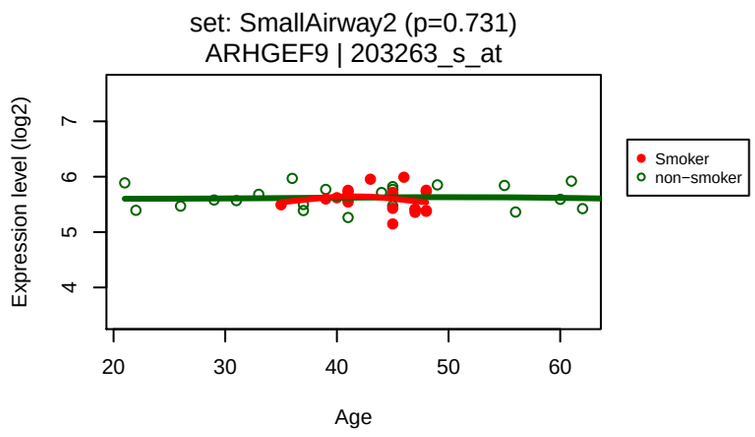
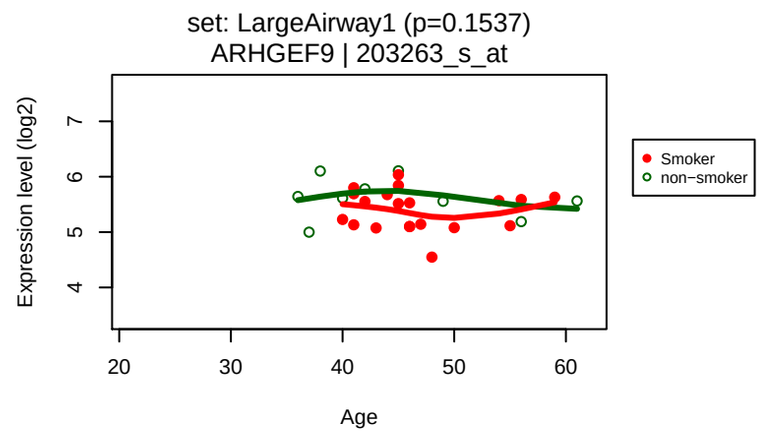

Gene expression of smokers and nonsmokers
gene: ARHGEF9 | 203263_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1555
p=0.731
p=0.3189
p=0.1537
p=0.3564
p=0.4428
p=0.4089
p=0.3449
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

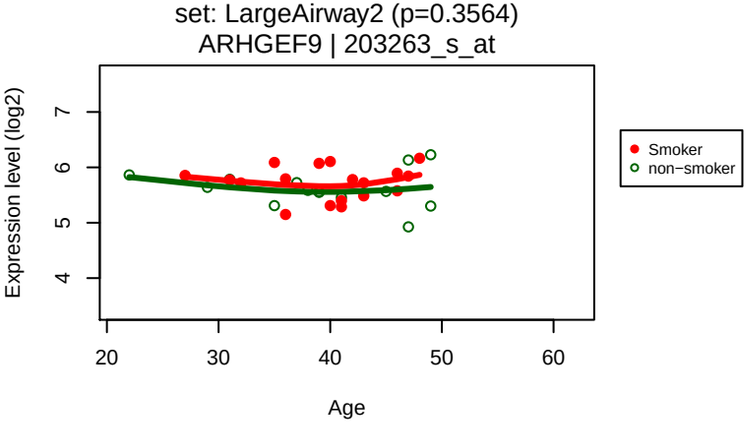

Gene expression ratio (smokers/nonsmokers)
gene: ARHGEF9 | 203263_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1555
p=0.731
p=0.3189
p=0.1537
p=0.3564
p=0.4428
p=0.4089
p=0.3449
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

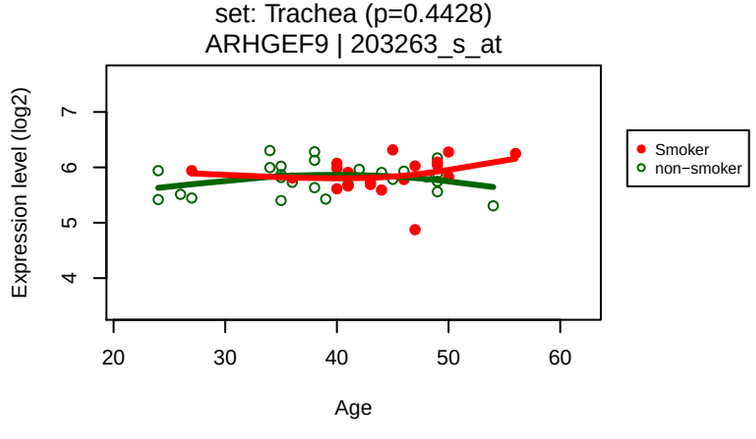

Gene expression ratio in Males
gene: ARHGEF9 | 203263_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4599
p=0.4965
p=0.8602
p=0.4362
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ARHGEF9 | 203263_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5541
p=0.2048
p=0.1762
p=0.7132
SmallAirway2
LargeAirway2
Trachea
Macrophage2

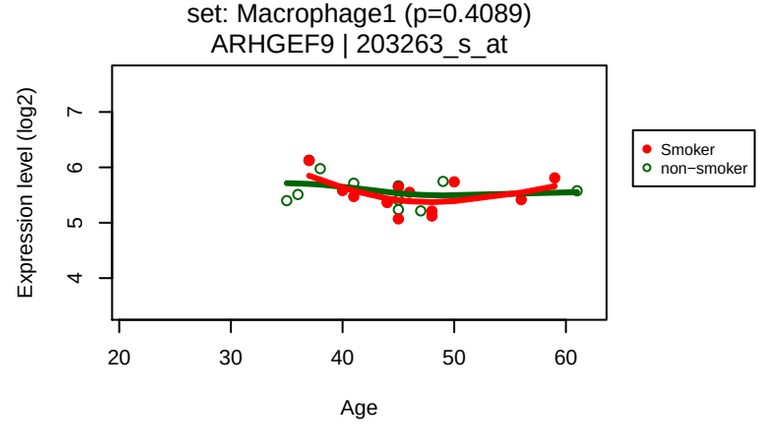

set: SmallAirway1 (p=0.1555)
ARHGEF9 | 203263_s_at
Expression level (log2)
Age
Smoker
non−smoker

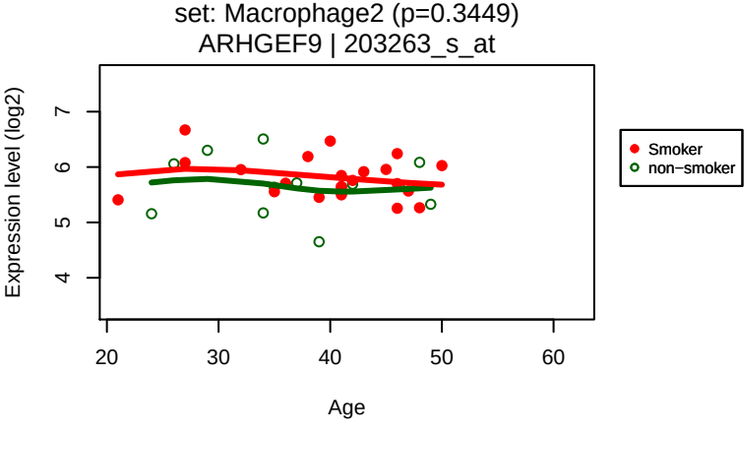

set: SmallAirway2 (p=0.731)
ARHGEF9 | 203263_s_at
Expression level (log2)
Age
Smoker
non−smoker

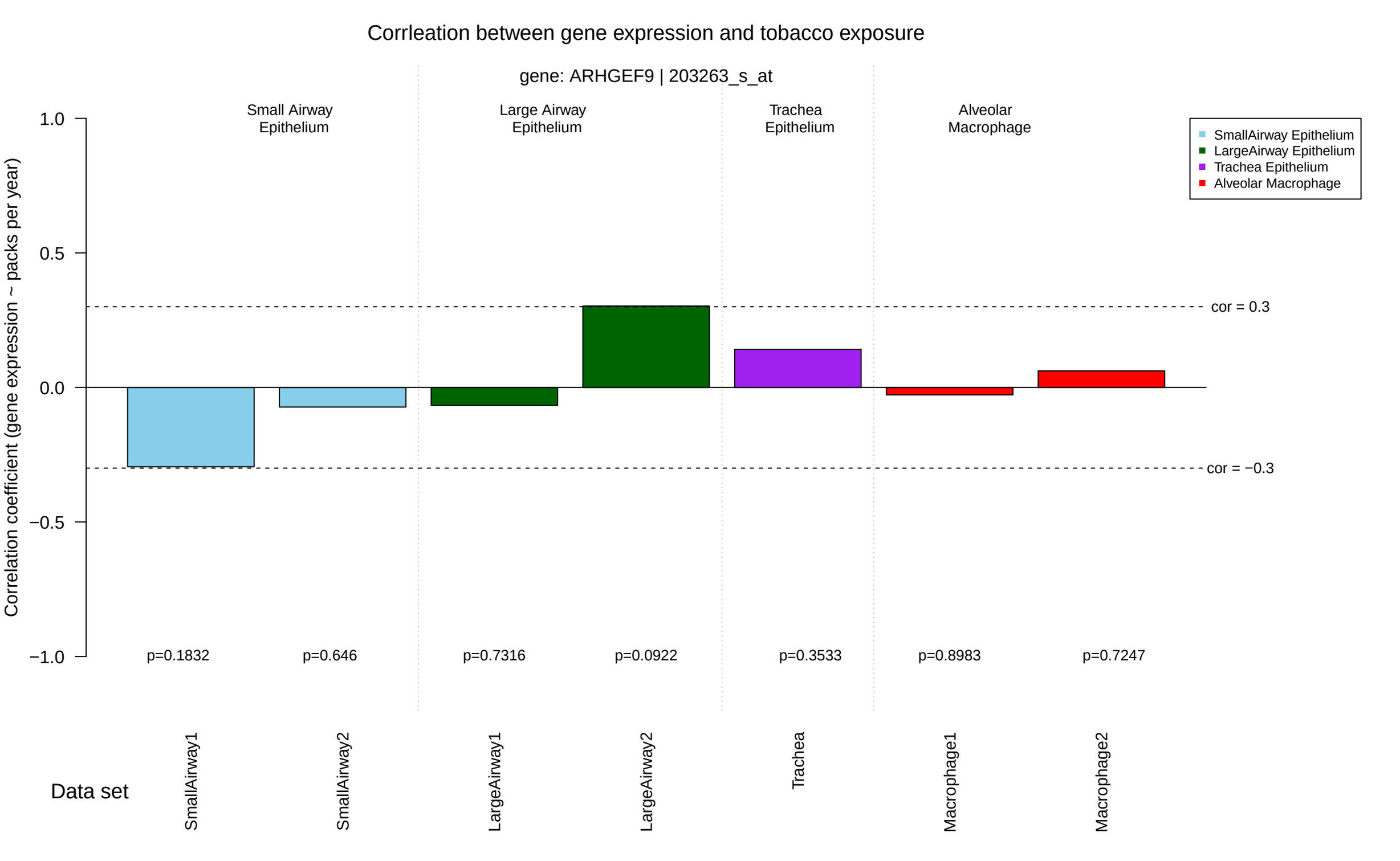

set: LargeAirway1 (p=0.1537)
ARHGEF9 | 203263_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3564)
ARHGEF9 | 203263_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4428)
ARHGEF9 | 203263_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4089)
ARHGEF9 | 203263_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3449)
ARHGEF9 | 203263_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ARHGEF9 | 203263_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1832
p=0.646
p=0.7316
p=0.0922
p=0.3533
p=0.8983
p=0.7247
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2